

Systems Analysis & Design

Semester 2026 - I

Workshop No. 4

Systems Analysis - Local Business Support Recovery

Group Seven

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1. Executive Summary

This workshop presents the development and analysis of simulation models for the Local Business Recovery Support System proposed for the Titan Plaza shopping mall environment. The objective of the simulations is to evaluate the operational behavior, performance, and resilience of the system under different conditions.

Two complementary approaches were implemented: a behavior-oriented simulation focused on adaptive interactions and emergent behavior, and a process-oriented simulation centered on operational workflows. The simulations were developed using SimPy, Tkinter, and Matplotlib.

The simulation model incorporates variables identified during previous workshops, including customer traffic variability, inventory dynamics, weather conditions, connectivity stability, and synchronization delays. Three scenarios were analyzed: baseline operation, optimization with predictive mechanisms, and failure conditions involving connectivity disruptions.

Results showed that the system maintains stable performance under normal conditions while demonstrating adaptive and resilient behavior during optimization and failure scenarios. The simulations also revealed emergent behaviors such as inventory fluctuations, bottlenecks during peak traffic periods, and the impact of synchronization delays on operational efficiency.

Overall, the simulation results validate the robustness and adaptability of the proposed system design while identifying opportunities for further optimization and future improvements.

2. System Simulation Model

2.1 Simulation Architecture

The simulation architecture was designed to represent the operational behavior of the Local Business Recovery Support System within the Titan Plaza environment. The model integrates key operational components such as customer traffic, inventory management, synchronization processes, promotional behavior, and connectivity conditions through an event-driven simulation structure. The architecture allows the interaction between multiple dynamic variables while supporting scenario-based experimentation and performance evaluation.

2.2 Behavioral and Process-Oriented Approaches

Two complementary simulation approaches were implemented. The behavior-oriented simulation focuses on adaptive interactions, system resilience, and emergent behaviors generated by changing operational conditions such as peak customer traffic, weather variability, and connectivity failures. In parallel, the process-oriented simulation models the sequential operational workflow of the system, including customer arrivals, sales processing, inventory updates, alert generation, and synchronization activities. Together, both approaches provide a broader understanding of system performance and operational dynamics.

2.3 Technologies and Implementation Decisions

The simulation was implemented using SimPy for event-based modeling, Tkinter for graphical visualization, and Matplotlib for performance analysis and chart generation. Several implementation decisions were based on operational factors identified during previous workshops, including customer traffic variability, weather conditions, inventory fluctuations, and connectivity constraints. A modular implementation structure was adopted to improve maintainability, scalability, and future extensibility of the simulation environment.

3. Experimental Design

3.1 Baseline Scenario

The baseline scenario represents normal operational conditions within the Titan Plaza environment. Customer traffic follows standard daily and weekly variations, inventory levels operate under regular restocking behavior, and system connectivity remains stable throughout the simulation. This scenario was used as the reference point for evaluating system performance and operational stability.

3.2 Optimization Scenario

The optimization scenario evaluates the effect of adaptive system mechanisms such as predictive restocking and dynamic promotional behavior. Under this scenario, the system reacts more efficiently to inventory depletion and customer demand variations, improving operational stability and reducing stock shortages during peak traffic periods.

3.3 Failure Condition Scenario

The failure condition scenario simulates connectivity disruptions and synchronization delays affecting the operational environment. During this scenario, the system operates with intermittent offline conditions, delayed synchronization processes, and reduced operational efficiency. This scenario was designed to evaluate system resilience, fault tolerance, and recovery behavior under adverse conditions.

3.4 Parameter Configuration and Testing Strategy

Simulation parameters were based on operational conditions identified during previous workshops (workshop 1 mainly), including customer traffic variability, weather effects, inventory dynamics, and connectivity constraints. Each scenario was executed multiple times to observe consistent behavioral patterns and compare system performance metrics such as sales activity, inventory stability, synchronization delays, and response times.

4. Results and Analysis

4.1 Performance Metrics and Statistical Results

The simulation generated multiple operational metrics to evaluate system performance under different scenarios. Key metrics included customer traffic behavior, sales activity, inventory fluctuations, response times, synchronization delays, and alert frequency. The collected data allowed the identification of operational trends, system stability levels, and performance differences between scenarios.

4.2 Behavioral Observations

The simulation results revealed different system behaviors in each of the evaluated scenarios.

In the baseline scenario, the system exhibited stable and predictable behavior. Sales increased steadily, inventory levels followed regular replenishment cycles, and response times remained within acceptable limits. This demonstrates that the system can operate reliably under normal conditions.

In the optimization scenario, a significant improvement in overall performance was observed. Thanks to predictive replenishment mechanisms and dynamic promotions, sales increased considerably, and response times were reduced. Although inventory fluctuated more frequently due to increased demand, the system maintained operational stability and continuous synchronization without delays.

On the other hand, the failure scenario revealed negative effects caused by connectivity and synchronization problems. Sales grew more slowly, inventory levels exhibited irregular behavior, and delays appeared that affected system efficiency.

A common pattern was identified in all three scenarios: Saturdays register the highest number of visitors, but this does not always translate into a proportional increase in buyers.

4.3 Comparative Scenario Analysis

Comparing the three scenarios revealed significant differences in the system's performance, efficiency, and responsiveness:

- In terms of sales, the optimization scenario achieved the best results, reaching approximately 1,200 cumulative sales over the 100 simulation days. In comparison, the baseline scenario registered around 510 sales, while the failure scenario reached approximately 420 (these results demonstrate the positive impact of predictive mechanisms and dynamic promotions)
- Regarding inventory management, the baseline scenario exhibited stable and predictable behavior. The optimization scenario showed more frequent replenishment cycles due to increased demand. Conversely, the failure scenario demonstrated greater difficulty in replenishing stock
- Synchronization was one of the aspects with the greatest differences between scenarios. The optimization scenario showed no delays, while the failure scenario experienced frequent delays that affected the overall system operation.
- Finally, visitor behavior was similar across all three scenarios, with Saturdays remaining the busiest day. However, the optimization scenario converted a higher proportion of visitors into buyers, while the failure scenario had the lowest conversion rates

5. Design Validation and Complexity Insights

5.1 Validation of Previous Design Decisions

1. The simulation results support the main design decisions proposed during Workshops confirming that the system is capable of addressing the problems and needs identified during the analysis phase.

2. One of the most important decisions was the implementation of synchronization mechanisms to address connectivity issues.

- The failure scenario demonstrated that connection interruptions can negatively affect parts of the system.
- Conversely, the optimization scenario maintained stable synchronization.

3. Furthermore, the modular architecture proposed in the previous workshops proved to be a sound decision, as it allowed the different system components to work in an integrated manner while maintaining flexibility and scalability.

4. On the other hand, the results support the phased implementation strategy proposed in Workshop #3.

5.2 Emergent Behaviors and Dynamic Interactions

During the simulations, several emergent behaviors were identified as a result of the interaction between the different system components:

One of the most relevant behaviors was observed in the optimization scenario. The combination of predictive replenishment mechanisms and dynamic promotions generated a significant increase in sales, achieving results far superior to those of the baseline scenario.

An important emergent behavior was also evident in the failure scenario. Connectivity problems caused delays in data synchronization, which affected the generation of inventory alerts and hindered replenishment processes.

Another pattern observed was the cyclical behavior of inventory. In all three scenarios, after an initial adjustment period, inventory levels tended to stabilize through repetitive cycles of consumption and replenishment.

Taken together, these results demonstrate that the interactions between the different components generate effects that go beyond their individual functioning, highlighting the importance of analyzing the system as an integrated whole and not merely as isolated elements.

5.3 Bottlenecks, Instabilities, and Sensitivity Effects

The simulation results identified several factors that limit the system's performance and could affect its operation in a real-world environment.

1. The main bottleneck identified was data synchronization. In the failure scenario, connectivity problems caused delays that affected information updates, inventory control, and sales processing.

(This demonstrates that the system is highly sensitive to connection interruptions.)

2. Another significant bottleneck was the discrepancy between the number of visitors and the number of actual buyers. Although Saturdays saw the highest traffic in all scenarios, conversion rates remained relatively low.

(This indicates that promotional and recommendation tools are essential to incentivize purchases.)

3. The rate at which inventory depleted during periods of high demand was also identified as a critical factor. In the optimization scenario, increased sales led to more frequent replenishment cycles, and at times, inventory levels approached critical values before being restocked.

(This suggests the need to adjust replenishment parameters to respond more efficiently to sustained increases in demand.)

4. Sensitivity effects, connectivity was found to be the variable with the greatest impact on overall system performance. The differences between the optimized and failure scenarios demonstrate that small variations in connection quality can produce significant changes in operation.

(Proper configuration will be key to successful system implementation.)

6. Recommendations

Based on the results obtained during the simulations, several recommendations are proposed to improve the system's performance and reliability.

1. **Implement strategies that allow for more efficient information storage and synchronization.**

First, it is recommended to strengthen synchronization mechanisms, as the failure scenarios demonstrated that connectivity problems have a significant impact on sales, inventory control, and the overall operation of the system.

2. **Adjust inventory replenishment policies.**

The use of predictive mechanisms would allow for anticipating periods of high demand and avoiding stockouts, especially during peak visitor traffic days.

3. **Promotions and recommendations could be implemented in the early stages of the system.**

The results showed that these tools have a positive impact on sales and on converting visitors into buyers, so their early adoption would increase the value generated for merchants.

4. Design specific promotional campaigns for weekends

Finally, it is recommended to run campaigns, especially on Saturdays, as simulations showed that this is the day with the highest number of visitors. Leveraging this influx of people through personalized offers and recommendations could help improve conversion rates and increase sales for local businesses.

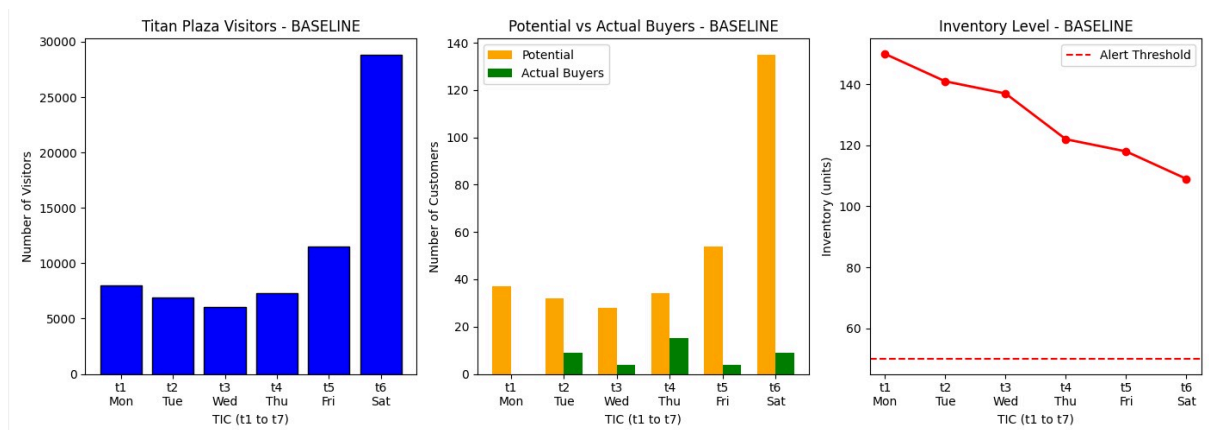
7. Visual Results and Analysis

This section presents the graphical outputs generated by both simulation approaches across the three evaluated scenarios: Baseline, Optimization, and Failure. The visualizations are organized by scenario to facilitate direct comparison of system behavior under different operational conditions.

Baseline Scenario

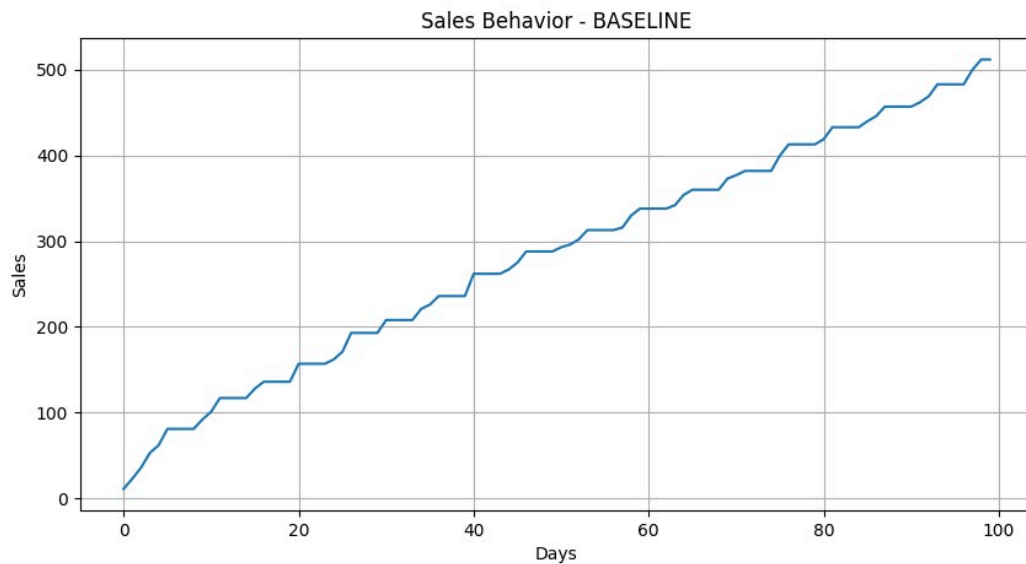
Titan Plaza Visitors, Potential VS Actual Buyers, inventory level. (Process-oriented simulation)

Visitor traffic peaks significantly on Saturday, reaching approximately 28,000 visitors. Despite this, the gap between potential and actual buyers remains wide across all days, with conversion rates staying low throughout the week. Inventory declines steadily without reaching the alert threshold.



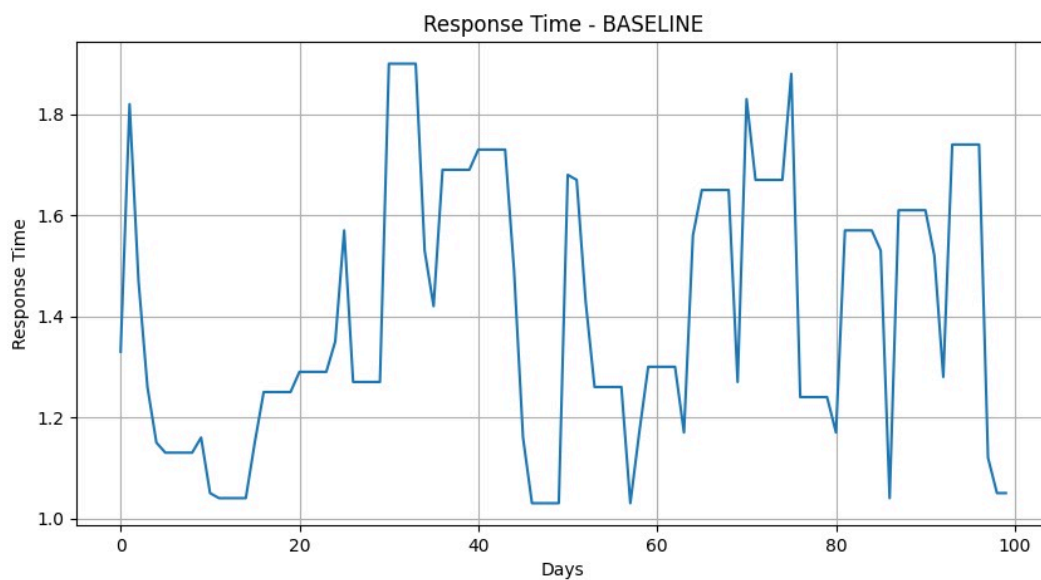
Sales Behavior. (Behavior-oriented simulation)

Cumulative sales grow steadily over 100 days, reaching approximately 510 units. The growth pattern is consistent and linear, reflecting stable but modest sales activity under normal operating conditions.



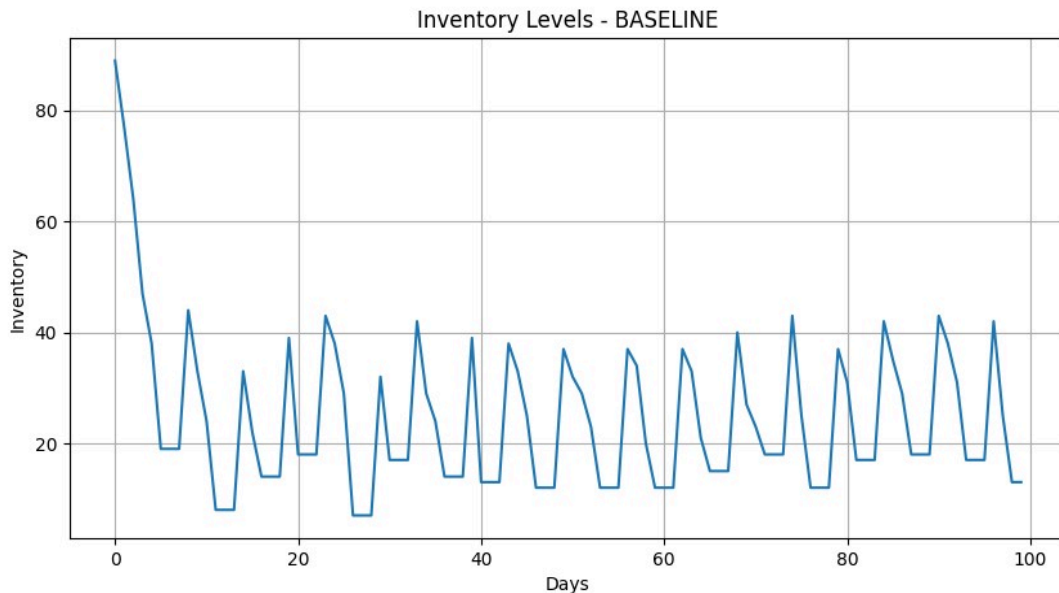
Response Time. (Behavior-oriented simulation)

Response time fluctuates between 1.0 and 1.9 throughout the simulation period, showing irregular variability without a clear trend. No critical peaks are observed, indicating acceptable system performance under normal conditions.



Inventory Levels. (Behavior-oriented simulation)

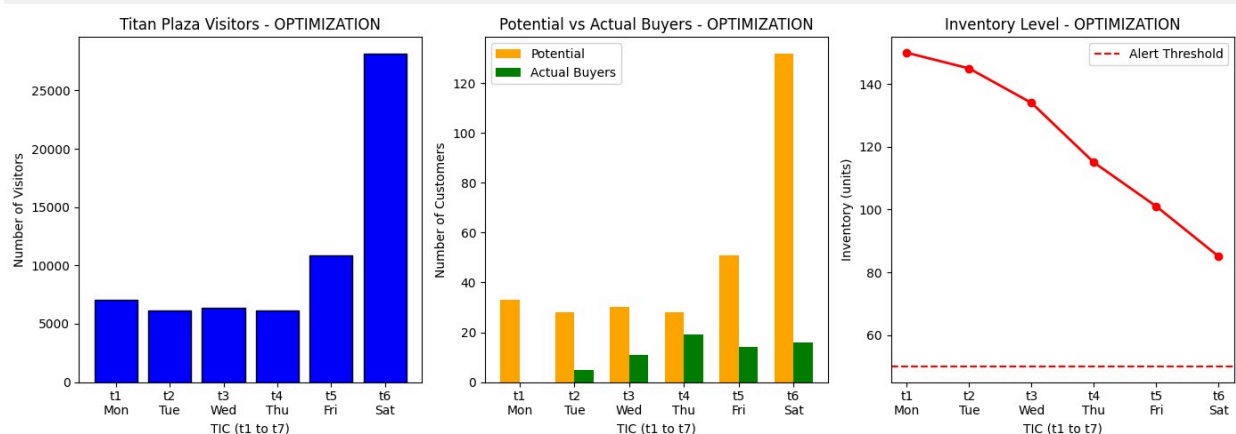
Inventory follows a recurring depletion and restocking cycle, stabilizing between 15 and 45 units after an initial drop from approximately 90. The pattern is consistent and controlled, reflecting regular inventory management behavior.



Optimization Scenario

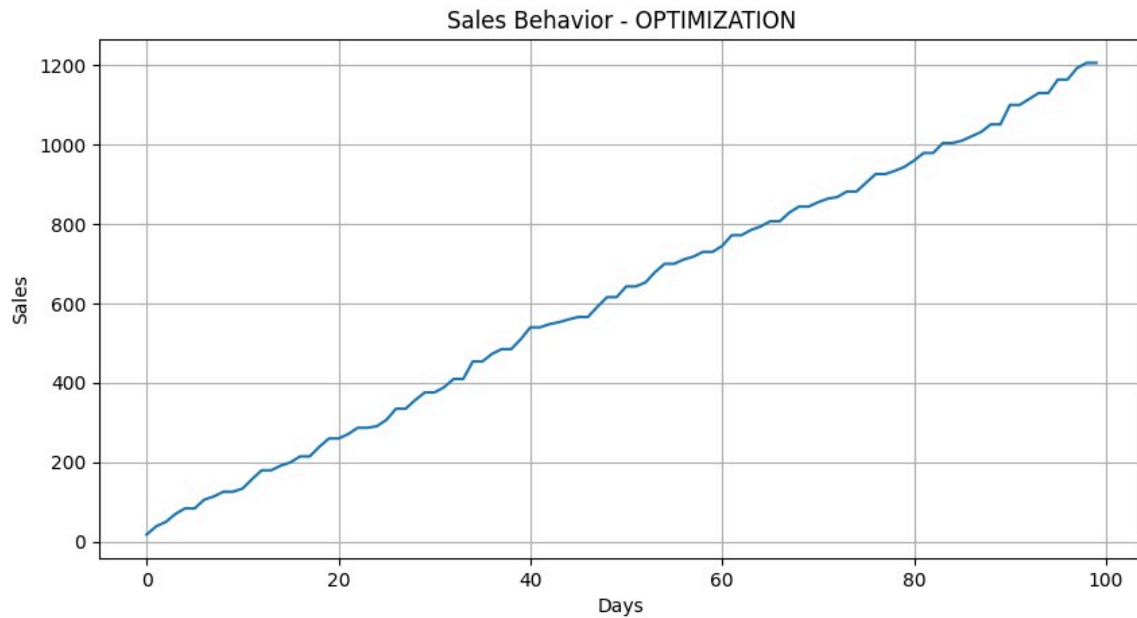
Titan Plaza Visitors, Potential vs Actual Buyers and Inventory Level. (Process-oriented simulation)

Visitor traffic follows a similar weekly pattern with Saturday peaks near 28,000. Notably, actual buyer conversion improves compared to the baseline, particularly on Thursday and Saturday. Inventory depletion is more pronounced, approaching the alert threshold by the end of the week.



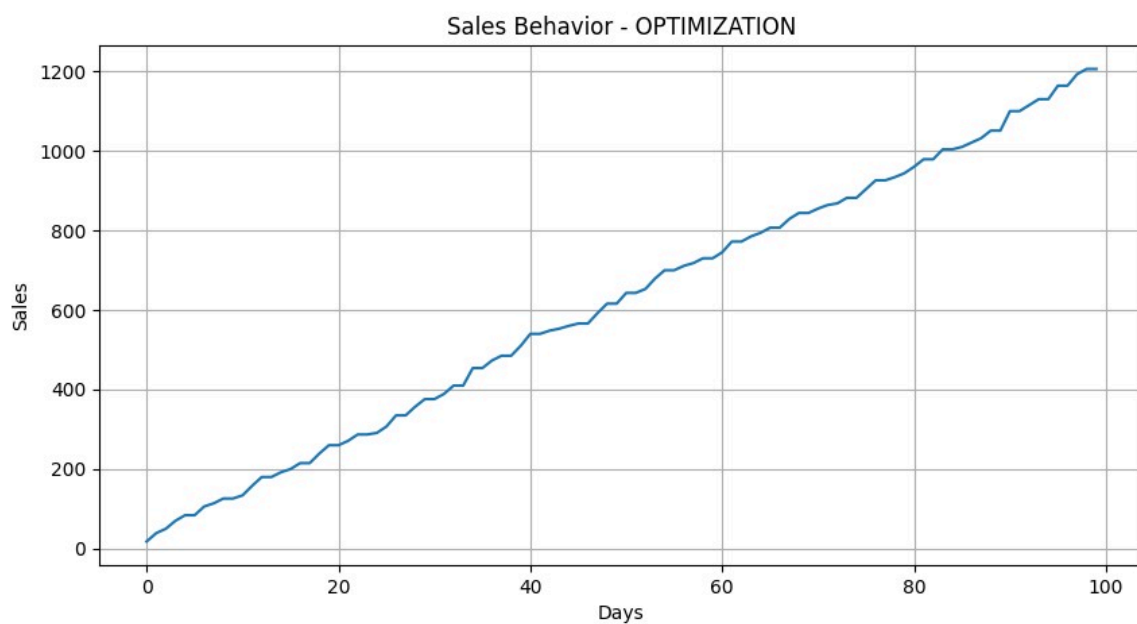
Sales behavior. (Behavior-oriented simulation)

Cumulative sales reach approximately 1,200 units over 100 days, more than double the baseline result. The growth curve is steeper and more consistent, reflecting the positive impact of predictive restocking and dynamic promotional mechanisms.



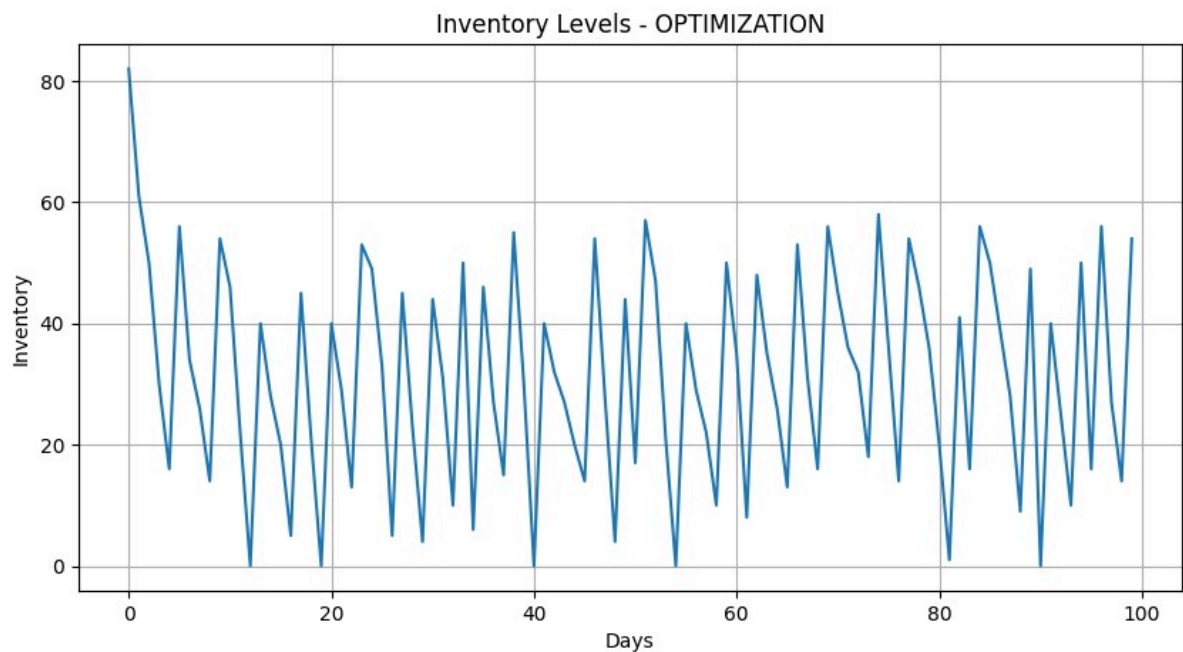
Response time (behavior-oriented simulation)

Response time ranges between 0.6 and 1.5, showing greater variability than the baseline but with lower average values. The reduction in response time reflects improved system efficiency under the optimization scenario.



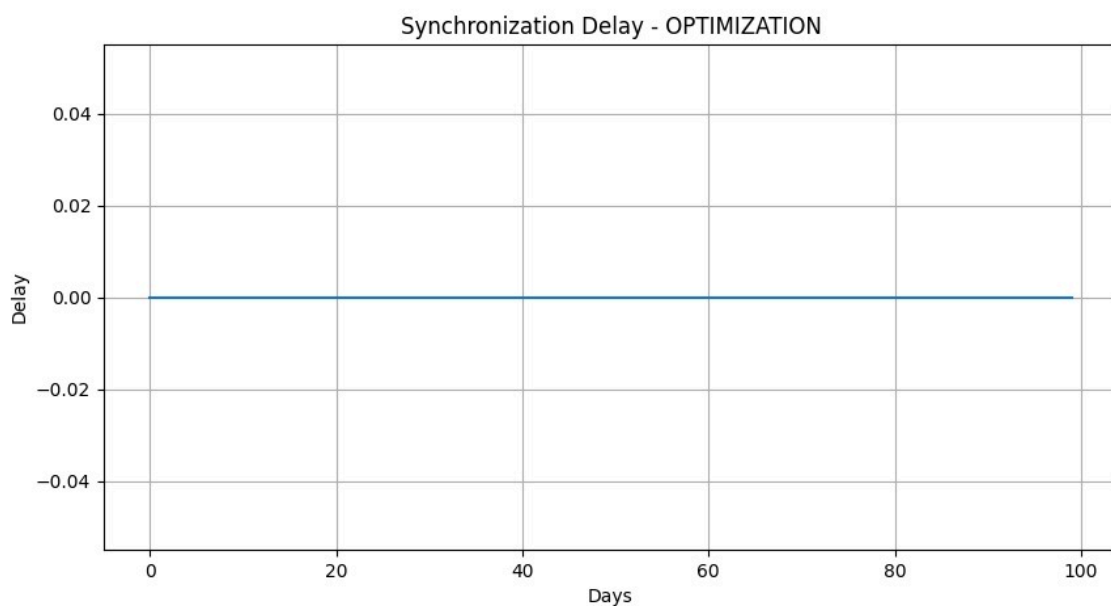
Inventory levels (behavior simulation)

Inventory oscillates more frequently between 0 and 58 units, indicating higher sales activity and more aggressive restocking cycles. The pattern reflects a more dynamic inventory environment driven by increased demand.



Synchronization Delay. (behavior-oriented simulation)

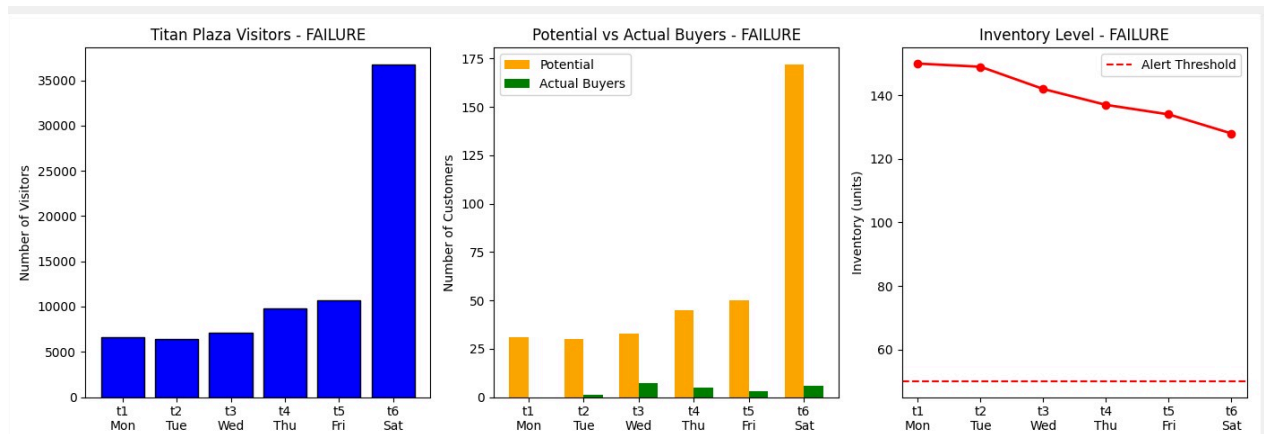
Synchronization delay remains consistently at zero throughout the entire simulation period, confirming that under optimized conditions the system maintains full connectivity and real-time data synchronization without interruptions.



Failure Scenario

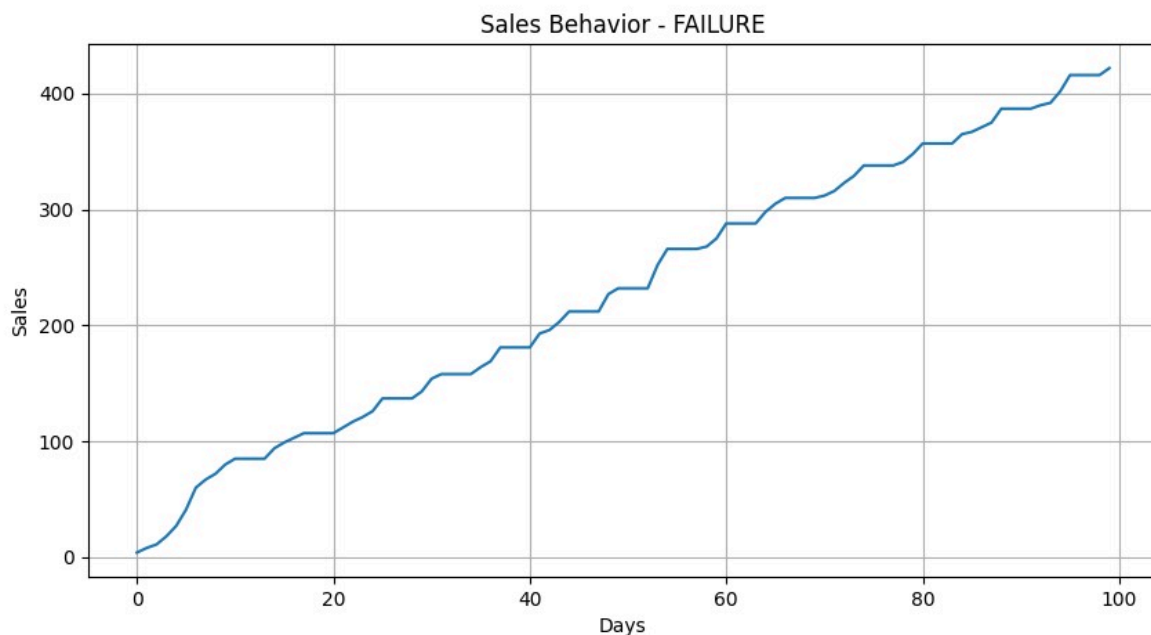
Titan Plaza Visitors, Potential vs Actual Buyers, and Inventory Level (Process-oriented simulation)

Visitor traffic spikes dramatically on Saturday, exceeding 36,000 visitors — the highest value across all scenarios. However, actual buyer conversion collapses almost entirely, with near-zero conversions on most days. Inventory continues declining without triggering effective restocking responses.



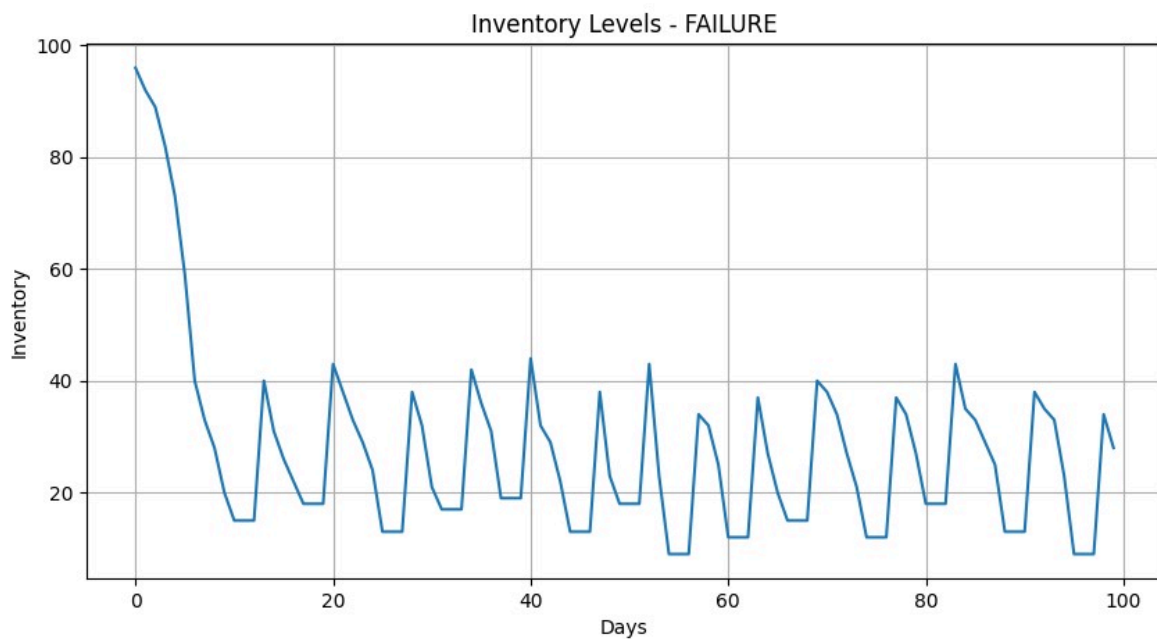
Sales behavior. (behavior-oriented simulation)

Cumulative sales reach only approximately 420 units over 100 days, the lowest result across all three scenarios. The growth curve is irregular and slower, reflecting the negative impact of connectivity disruptions on sales processing and system responsiveness.



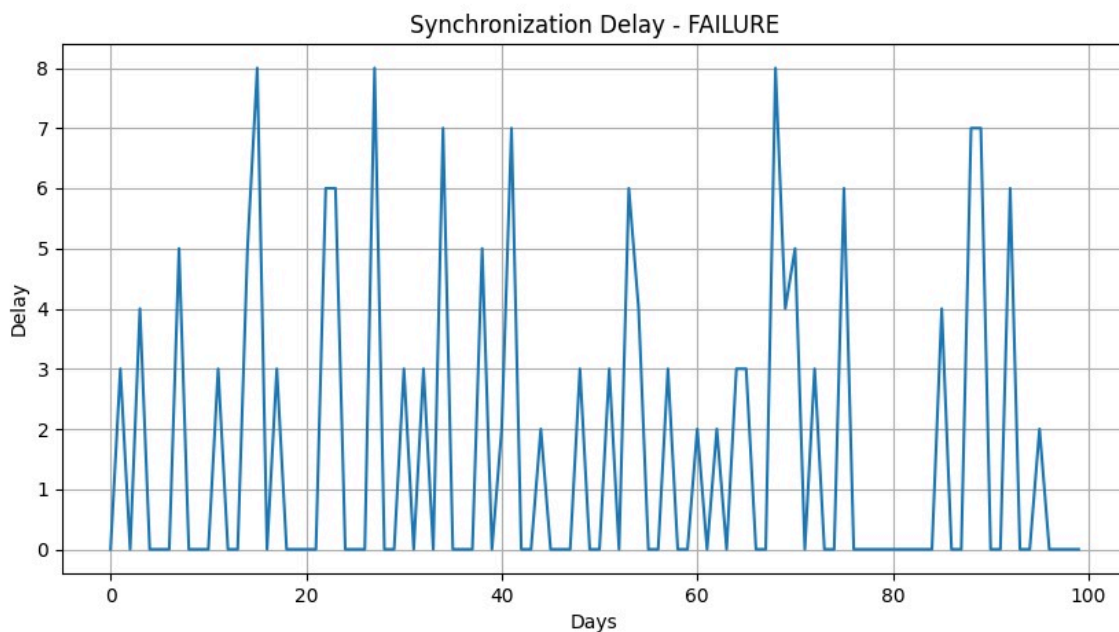
Inventory levels (behavior-oriented simulation)

Inventory follows an erratic depletion pattern, dropping to critically low levels near 10 units on several occasions. The restocking cycles are inconsistent, reflecting the impact of synchronization failures on inventory management reliability.



Synchronization Delay (Behavior-oriented simulation)

Synchronization delay fluctuates irregularly between 0 and 8 units throughout the simulation, with frequent spikes. This pattern confirms that connectivity disruptions directly affect data synchronization, creating operational instability and delayed system responses.



8. Conclusions

The simulations showed that the system works as expected. When optimization features are active, sales more than doubled compared to baseline conditions, which confirms that tools like predictive restocking and promotions make a real difference for small businesses.

The failure scenario made it clear that connectivity is the system's biggest vulnerability. When synchronization breaks down, everything else suffers, like inventory management, sales processing, and response times all degrade together. This makes offline synchronization one of the most important things to get right before deployment.

One pattern that appeared consistently across all three scenarios was the gap between visitors and actual buyers on Saturdays. High foot traffic does not automatically mean more sales, which reinforces why the platform's engagement features are not just nice to have, its necessary.

After four workshops, the project has gone from observing a real problem at Titan Plaza to designing, planning, and now simulating a solution that holds up under different conditions.

9. References

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